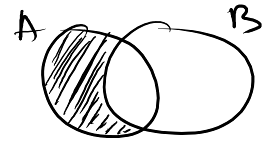
Sunday, September 22, 2024 7:41 AM

Given $A \in \mathcal{F}$, $B \in \mathcal{F}$, $\mathcal{F} \to \sigma$-algebra

$$\begin{cases} \phi \in \mathcal{F} \\ A \cup B \in \mathcal{F} \\ A^c, B^c \in \mathcal{F} \end{cases}$$



① $A \cap B \in \mathcal{F}$

$A^c \in \mathcal{F}$, $B^c \in \mathcal{F}$

$(A^c \cup B^c) \in \mathcal{F}$

$(A^c \cup B^c)^c \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$
(demorgan's law)

② $A - B \in \mathcal{F}$

$A \cap \boxed{B^c} \in \mathcal{F}$

$\mathbb{E}[X|Y=0] = ?$

ω	1	2	3	4	5	6
$P(\omega)$	1/6	1/6	1/6	1/6	1/6	1/6
X	0	1	0	1	0	1
Y	0	1	1	0	1	0

$$\begin{array}{c|cc} X \backslash Y & 0 & 1 \\ \hline 0 & 1/6 & 2/6 \\ 1 & 2/6 & 1/6 \end{array}$$

$P(X=x, Y=y)$ (Joint Pmf) $P_{X,Y}$

X	0	1
	1/2	1/2

P_X (Marginal) $P(X=x)$

Y	0	1
	1/2	1/2

P_Y (Marginal) $P(Y=y)$

$$\mathbb{E}[X] = 0 \times \mathbb{P}(X=0) + 1 \times \mathbb{P}(X=1)$$
$$= 0 \times \tfrac{1}{2} + 1 \times \tfrac{1}{2} = \tfrac{1}{2}$$

$$\mathbb{E}[X|Y=1] = 0 \times \mathbb{P}(X=0|Y=1) + 1 \times \mathbb{P}(X=1|Y=1)$$
$$= 0 + \frac{\mathbb{P}(X=1, Y=1)}{\mathbb{P}(Y=1)} = \frac{1/6}{1/2} = \tfrac{1}{3}$$

$$\mathbb{E}[X|Y=0] = 0 \times \mathbb{P}(X=0|Y=0) + 1 \times \mathbb{P}(X=1|Y=0)$$
$$= 0 + \frac{\mathbb{P}(X=1, Y=0)}{\mathbb{P}(Y=0)} = \frac{2/6}{1/2} = \tfrac{2}{3}$$

$\boxed{\mathbb{E}[X|Y]} \rightarrow \tfrac{1}{3}, \tfrac{1}{2}$

$\boxed{\mathbb{E}[X|Y]} \rightarrow \tfrac{2}{3}, \tfrac{1}{2}$

(2) joint density $f_{XY}(x,y) = x^2 + \frac{4y}{3}$, $0 \le x \le 1$, $0 \le y \le 1$

(i) ~~proper~~ proper density ??

$$\int_0^1 \int_0^1 f_{XY}(x,y)\,dx\,dy = 1$$

$f_Y(y) = ?$ $\quad f_Y(y) = \int_0^1 f_{XY}(x,y)\,dx = \int_0^1 \left(x^2 + \frac{4y}{3}\right) dx$

$$= \left[\frac{x^3}{3} + \frac{4yx}{3}\right]_0^1 = \frac{1}{3} + \frac{4y}{3} = \frac{1+4y}{3}$$

$$\int_0^1 f_Y(y)\,dy = \int_0^1 \left(\frac{1+4y}{3}\right) dy = \left[\frac{y+2y^2}{3}\right]_0^1 = 1 - 0 = 1$$

$$\mathbb{E}[X|Y=y] = \int_0^1 x f_{X|Y}(x|y)\,dx$$

$$= \int_0^1 x \left(\frac{x^2 + 4y/3}{\frac{1+4y}{3}}\right) dx = \int_0^1 x\left(\frac{3x^2+4y}{1+4y}\right) dx$$

$$= \int_0^1 \frac{3x^3 + 4yx}{1+4y}\,dx$$

$$= \frac{1}{1+4y} \cdot \left[\frac{3x^4}{4} + 2yx^2\right]_0^1$$

$Y=y$

$$= \frac{3/4 + 2y}{1+4y} = \frac{3+8y}{4+16y}$$

$\mathbb{E}[X|Y]$ → ( ), $Y=1$

$\mathbb{E}[X|Y \ldots]$ $\quad 3+8y$

$\mathbb{E}[X|Y]$ $\rightarrow$ ( ), $Y=1$
$\rightarrow$ ( ), $Y=2$

r.v

$\rightarrow$ $\mathbb{E}[X|Y=y] = \dfrac{3+8y}{4+16y}$

$\mathbb{E}[X|Y] = \boxed{\dfrac{3+8Y}{4+16Y}}$

$$X_i \sim (\mu_X, \sigma_X^2)$$
$$N \sim (\mu_N, \sigma_N^2)$$

$$Y = X_1 + X_2 + \cdots X_N$$

$$\mathbb{E}[Y \mid N = n] = \mathbb{E}[X_1 + X_2 + \cdots X_N \mid N = n]$$
$$= \mathbb{E}[X_1 + X_2 + \cdots X_n]$$
$$= n\mu_X$$

$$\mathbb{E}[Y \mid N] = N\mu_X \quad \text{-- (random variable)}$$

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid N]] = \mathbb{E}[N \cdot \mu_X]$$
$$= \mu_X \cdot \mu_N$$

$$\text{var}(Y) = \mathbb{E}[\text{var}(Y \mid N)] + \text{var}[\mathbb{E}[Y \mid N]]$$